

Dear Sir or Madam,

I am applying for the Senior Research Engineer, Data Engine role because my work has focused on turning advanced research ideas into reliable, high-performance systems that generate and process large-scale scientific data.

At QC Design, I led the development of the company's core quantum simulation and analysis platform used directly by customers and researchers. A central part of my role was prototyping and productionizing new simulators, error-correction codes, and optimization methods based on recent research papers. I regularly translated theoretical models into efficient C++ and Python implementations, verified their correctness against the literature, and integrated them into a maintainable codebase. By designing custom CPython bindings and performance-critical C++ components, I achieved over $10,000\times$ speedups, making large experimental studies practical in production.

In parallel, I built testing and benchmarking pipelines to validate numerical accuracy and reproducibility, and I used cloud infrastructure to run long-duration simulation campaigns that produced large datasets for analysis. I worked closely with researchers to investigate unexpected results, refine models, and iterate on algorithms until they were scientifically correct and reliable in large-scale runs.

Earlier, as CTO of WD AI, I gained experience deploying ML services and data pipelines at scale using Docker, Kubernetes, and message queues, which complements my research background with strong production experience.

Across my work, I have focused on building infrastructure that enables researchers to run complex experiments, trust their data, and move faster. I am excited about applying this experience to Intrinsic's mission of advancing intelligent robotics through scalable data and ML systems.

Yours truly,

Ivan Ogloblin